# Opgeloste examenvragen Computer Networks

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## Application Layer

## Gesloten boek

**1) HTTP**

**a) Wat is het verschil tussen HTTP 1.0 en 1.1?**

1.0: moet een nieuwe TCP-connection openen voor elk object (=> inefficiënt want elke webpagina bevat gemiddeld 40 embedded objects + TCP-congestion-algoritme verhoogt de snelheid geleidelijk aan met de tijd, dus transfer rate nieuwe connecties is trager), 1.1 maakt gebruik van persistent TCP connections, daarboven maakt 1.1 gebruik van pipelining: het download elk embedded object zodra geweten is dat het nodig is ipv wachten tot de transfer van zijn parent voltooid is.
1.1 vereist HOST-header, 1.0 niet
1.1 heeft OPTIONS-method, 1.0 niet.

**b) wanneer is dit performantievoordeel het grootst?**

-Langdurige connecties
-Veel embedded objects

**2) DHCP**

**a) Leg uit hoe DHCP geïmplementeerd wordt, je kan kiezen om het via woorden uit te leggen of via een getekend schema.**

DHCP: provides automated assignment of IP-addresses, the IP-addresses are assigned on demand. The addresses are leased for a specific period, which is tailored to the environment: shorter leases more efficiently allocate space, longer leases lower load on the DHCP server.

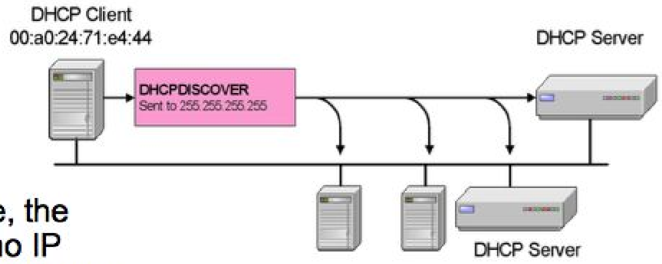
After assignment of an IP address, hosts will perform duplicate address detection, if the client detects a duplicate address, it refuses the offered IP address with a DHCPDECLINE message. Leases are renewed when 50% has expired. If DHCP server sends DHCPNACK, then address is released.

(See images below for discovery, address acquiring and release)

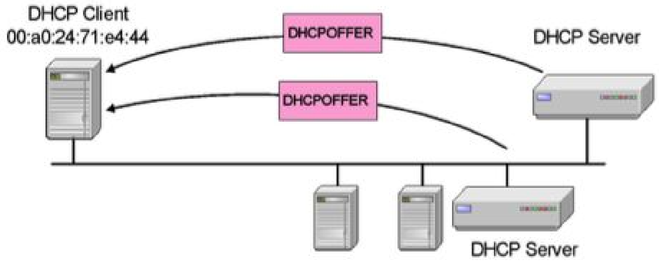
**b) Kan DHCP via TCP geïmplementeerd worden?**

No, since the client doesn't know the IP address of the DHCP server it will broadcast a UDP message. The server will also broadcast a UDP message since the client has no IP address yet.

# DHCP Discovery

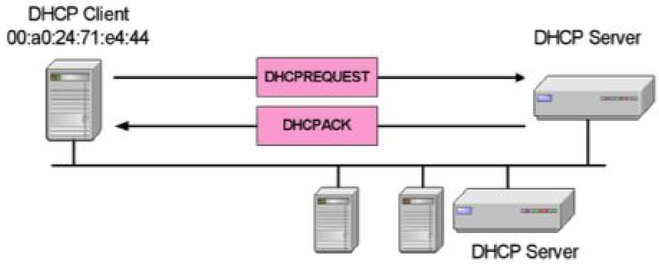



**Note:** at this stage, the DHCP client has no IP address. All messages are broadcast.

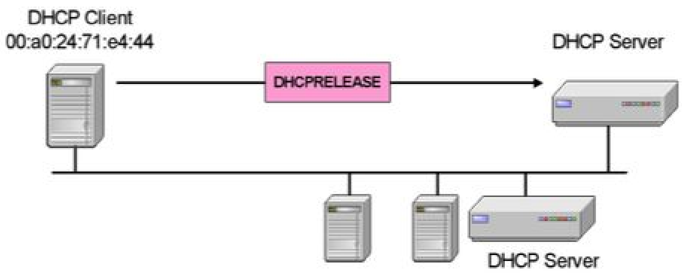



# Acquiring an Address

**Note:** the client now has an IP address and all further communication is via IP.




# Releasing an Address



**3) TOR**

**a) Welke informatie heeft een entry routing node, een ending routing node en een intermediate routing node?**

- **entry**: heeft een lijst van TOR routers, kiest dan een random pad en encrypt het bericht meerdere keren in volgorde, het heeft dus ook alle public keys van de onion routers waarlangs hij het bericht stuurt.
  Pas op: Onion Proxy ≠ Onion Router
  entry ROUTING node, weet die niet gewoon hetzelfde als een intermediate?
- **ending**: kent de vorige hop en kent de inhoud van het bericht, deze is echter meestal nog beveiligd via TLS of HTTPS.
- **intermediate**: kent enkel de next hop en previous hop.

**b) Hoeveel intermediate nodes zijn er minstens nodig voor een veilige verbinding?**

Technisch gezien maar 1 intermediate nodig om veilig te zijn aangezien de exit node dan niet kan weten wie het bericht verstuurd heeft. Echter is de standaard in TOR om minstens 2 intermediate nodes (3 hops) te hebben aangezien als in het vorige geval een adversary de exit node bezit of bekijkt, ze weten welke andere node ze zouden moeten compromisen voor ze weten wie ge zijt.

**c) Heeft het zin om er veel meer bv. 10 te hebben?**

Nu komt automatisch de vraag, waarom standaard 3 en niet 4, 5, …? Wel, op een bepaald punt zorgt het toevoegen van extra hops voor extra latency en niet echt veel extra beveiliging. Daarbovenop zijn er veel aanvallen die toch geen fuck geven hoe lang ge aan het hoppen zijt binnen de Tor cloud anyway. Zie filmpje vanaf 10m39 voor een voorbeeld https://youtu.be/QRYzre4bf7I?t=10m39s
3 hops wordt dus algemeen genomen als een goede compromise.

**4) Gnutella**

**a) You are employed by a copyright enforcement agency that wishes to monitor Gnutella 0.4. List each type of message that you would monitor and what relevant information it would reveal about users of the Gnutella network.**

- Query: relevant to know which nodes request which files at any time
- Queryhit: relevant to know which node hosts which files

**b) You are asked to transition the monitoring software described in Part 1.2 to the Gnutella 0.6 network:**

- **How would you connect to the Gnutella 0.6 network to monitor it most effectively?**

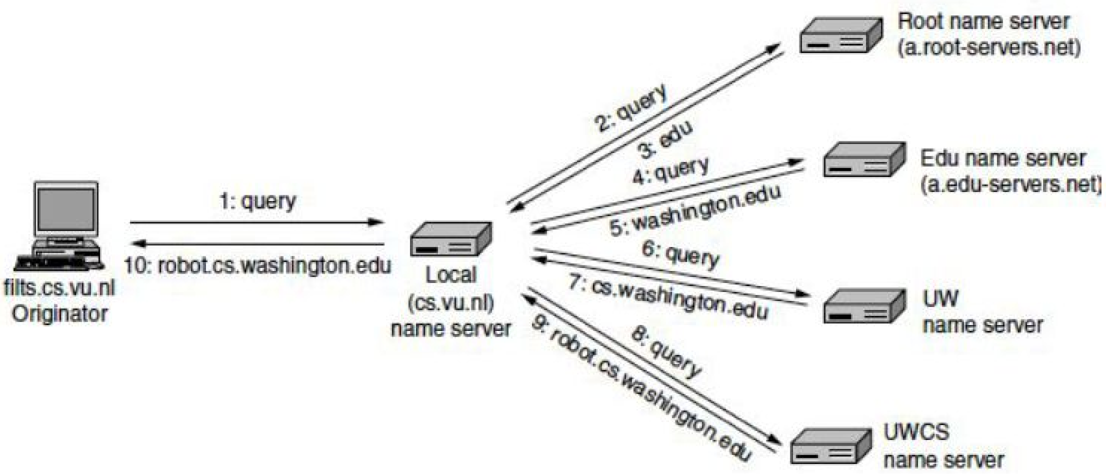
As an ultra-node, as this will intercept queries.

- **Do you expect to gather more or less data on Gnutella 0.6 compared to Gnutella 0.4?**

More, as more queries pass through these ultra-nodes (they only pass through these nodes), than did through the nodes in Gnutella 0.4

**5) DNS**

**a) Hoe werkt DNS? Leg uit en illustreer met tekening, duid ook aan welk deel recursief en welk iteratief is.**




Het recursieve element: de local-name-server geeft geen partiële info terug
Het iteratieve element: de root name server geeft enkel partiële informatie terug

**b) Stel dat er enkel een recursief deel is, hoe zou het er dan uitzien en wat zijn dan de voor-/nadelen?**
Als er enkel een recursief deel is, dan wil het zeggen dat alle andere name servers ook geen partiële informatie zouden teruggeven. Na de query naar de root name server in stap 1, zou de de root name server dus zelf query'en naar de TLD name server enzoverder, als het gezochte IP-adres gevonden is wordt het teruggestuurd via de verschillende name servers wat tot overhead leidt, hierdoor wordt het in de praktijk niet gedaan.

**c) Hoe zou je DNS testen met Telnet? Verklaar.**
Je kan proberen om een telnet verbinding te maken op het IP-adres van de local name server, echter zal de verbinding direct gesloten worden aangezien er niet geïnterageerd wordt op de verwachte manier met de name server. DNS is namelijk een binary protocol, dit betekent concreet dat je een DNS client program nodig hebt om met de name servers te kunnen interageren.

**d) Hoe zou je DNS dan wel kunnen testen?**
Manueel itereren via het dig command.

**6) Vergelijk en contrast de sustainability van Gnutella 0.4 en Chord voor het ondersteunen van de volgende toepassingen:**

**a) een betrouwbaar distributed file storage system**

- Gnutella 0.4 is inferior, as it cannot provide reliable access to all files. The limited broadcast (search) horizon limits this.
- Chord however, does guarantee access to all files, even when the nodes themselves are unreliable.

**b) file sharing system**

**c) distributed DNS implementation**

- Gnutella would encounter a higher overhead due to the broadcasting mechanism
- Gnutella wouldn't function well because the search horizon is limited (problem for DNS)

**d) video-conferencing system**

- Video-conferencing implies multiple simultaneous users, hence

## Open boek

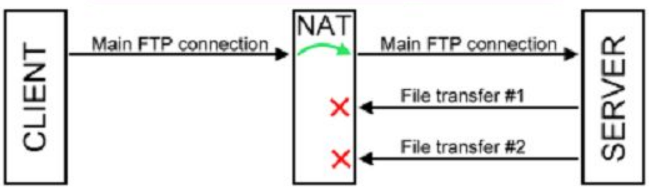
**1) FTP**

**a) Waarom gebruikt men verschillende verbindingen voor control en data-transfer?**

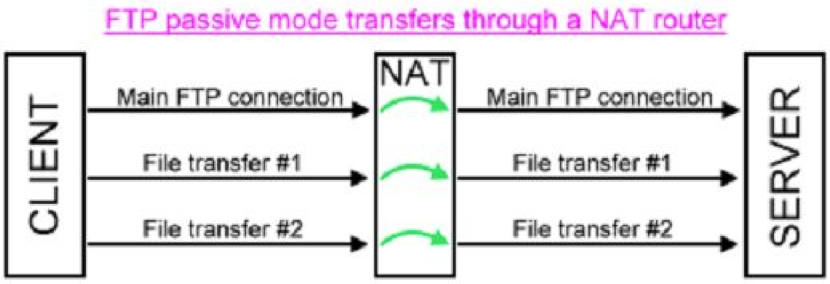
FTP actually uses two channels between client and server, the command and data channels, which are 2 separate TCP connections.
The command channel is for commands and responses while the data channel is for actually transferring files.
This separation of command information and data into separate channels a nifty way of being able to send commands to the server without having to wait for the current data transfer to finish. This is helpful for a subset of commands, such as quitting, aborting the current transfer, and getting the status.

**b) Wat is het probleem bij FTP met NAT?**

FTP active mode transfers through a NAT router




**c) Hoe lost passive mode dit op?**




In active mode, the client establishes the command channel but the server is responsible for establishing the data channel. This can actually be a problem if, for example, the client machine is protected by firewalls and will not allow unauthorised session requests from external parties.
In passive mode, the client establishes both channels. We already know it establishes the command channel in active mode and it does the same here. However, it then requests the server (on the command channel) to start listening on a port rather than trying to establish a connection back to the client.
As part of this, the server also returns to the client the port number it has selected to listen on, so that the client knows how to connect to it.
Once the client knows that, it can then successfully create the data channel and continue.

**2) Building Gnutella over Chord met text-based searching (dus hashen voor queries te doen gaat niet werken):**
**a) Wat vervangt Chord?**
Het query en queryhit systeem van gnutella?
Klopt denk ik. Chord implementeert een DHT (Distributed Hash Table) om zo makkelijk opzoekingen te doen (wie heeft welke file), dus keys (peer met queried file) mappen op values (queried file).
**b) Denk je dat Gnutella over Chord draaien een performantieverschil geeft (vergeleken met gewone Gnutella lijkt me)?**
Ja, elke node is bereikbaar en dus ook elk bestand, in tegenstelling tot gnutella waar niet alle nodes (en dus ook niet altijd alle bestanden) bereikbaar zijn door de search horizon. (search horizon: broadcast search messages hebben een TTL waardoor een broadcast search een gelimiteerd bereik heeft).
Chord schaalt met log(N), dus bij grote netwerken is er zeker een performantieverschil met Gnutella.
**c) Na file discovery kan je de file downloaden met een directe TCP-verbinding maar ook via de Chord-overlay. Geef een voordeel en nadeel aan 1 methode tegenover de andere.**

**3) Describe a high-level of Gnutella on top of Chord**
**a) Which protocol does Chord replace?**

**b) Give advantages and disadvantages concerning features, network overhead and anonymity.**

**4) a) What requirements does Chord place on its hash**

It needs to take a variable-length byte string as an argument and produce a highly random 160-bit number. We also need to be able to find *successor(key)* easily.

**b) What would be the effect if this requirement wasn't fulfilled?**

**5) Geef 3 applicaties die UDP gebruiken en leg uit waarom ze geen TCP gebruiken?**

- Wake on LAN
  a service that can wake computers from sleep mode when they receive a certain packet. No TCP because the computers are off, they have no IP address, the interaction is short (a binary switch) and network cards are simple, they should not need a full TCP stack.
- DHCP
  het eerste bericht dat de DHCP verstuurt, de DISCOVER, is een broadcast aangezien het IP-adres niet gekend is, en TCP geen broadcast messages ondersteunt. Daarom gebruiken we UDP.
- DNS
  Short messages, a three-way handshake for every query would cause too much overhead.

**6) Vergelijk Gnutella 0.4, 0.6 en Chord qua privacy**

In Gnutella 0.6 only ultra-peers participate in peer discovery, file discovery messages are only sent to leaf-nodes where they host a matching file. Monitoring ultra-peers (in Gnutella 0.6) will provide more information than monitoring ‘normal’ peers in both Gnutella 0.4 and 0.6.

# <u>Transport Layer</u>

## <u>Gesloten boek</u>

**1) The RTP protocol:**

**a) List the features of RTP, and why they are needed.**

- Each RTP package includes
  - a linearly increasing sequence number 1 higher than its predecessor. (We still need to know if packets went missing, but the action we take could be different: ignore, interpolate)
  - A timestamp. The difference relative to the first package might be interesting in regenerating the media content
  - A profile-specific marker (eg for start of frames)
  - Type of encoding algorithm used
- RTP packages are encapsulated within UDP segments
- RTP also supports extension headers for more advanced functionality.
- Real Time Control Protocol (RTCP) controls RTP streams. It provides feedback and control on delay, variation in delay, jitter (= the variability in delay), bandwidth, congestion.
  Provides synchronization of streams using clocks of different granularity.
  Provides human-readable naming of sources.

**b) List the features of TCP that are not in RTP, and why there are not in RTP.**

Flow control, ordering, congestion control, reliability because it’s implemented over UDP. This is so it provides a low overhead so it can be implementer on tiny devices.
In video-playback, retransmission of a frame isn’t very useful (it’s too late anyway)
We don’t necessarily need every frame anyway. We can interpolate (i.e. guess at the contents based upon the previous and following frames) or ignore.
Multicast is a critical tool for efficient media distribution.

**2) Describe the TCP flow control process. What is the difference with a standard selective repeat protocol?**

Flow control takes place through sliding windows: Sliding window protocols tackle both error and flow control.
Senders and receivers each maintain a window of messages for which no ACKs have been received. This is a sequence of message IDs, with a lower and upper bound. By configuring the window size we can control the flow of packets into the network, the size can be fixed or variable.
TCP **decouples** ACK from buffer allocation => receiver must independently ACK and advertise available buffer space.
Stop-and-wait: buffer 1 at sender and receiver, wait until ACK before sending next segment, if no ACK before timeout => resend. If bi-directional connections:

piggyback ACK’s in data segment. Simple, cheap but inefficient if high RTT compared to speed.
Go-back-N: Buffer W at sender, 1 at receiver, send W segments before requiring an ACK, sender has per-segment timer, higher ACK’s acknowledge all lower segments, better performance by parallelizing sending of packets (esp. if RTT is long and bandwidth high)
Selective repeat: W buffer at sender and receiver (=> packets can be received out of order), NACK for damaged or out of order segments, time-out cases individual resend (eliminates wasted retransmission of received packets)

TCP is similar to selective repeat because, when packets are lost due to congestion, the protocols do not require the sender to retransmit EVERY unACK'd packet sent by the sender. The sender just retransmits the oldest unACK'd packet. TCP is different from selective repeat because SR requires individual acknowledgement of each packet that was sent by the receiver; but rather than selectively ACKing every packet, TCP sends an ACK for the next packet that it is expecting (like Go Back N) and buffers the ones that it has received so far, even if they're out of order (like SR). That's why it can be considered a hybrid.

**3) OSPF uses UDP, BGP uses TCP. Give reasons why these design choices were made.**
OSPF makes use of the Link State Routing mechanism, which requires the distribution of the Link State Routing Tables, as implemented through flooding, for which broadcast is required.

The Border Gateway Protocol however, must only advertise routes to neighboring AS boundary routers, for which no broadcasting is required. As it maintains highly active links, the TCP connection overhead is negligible when compared to the reliability, congestion control and flow control it offers.

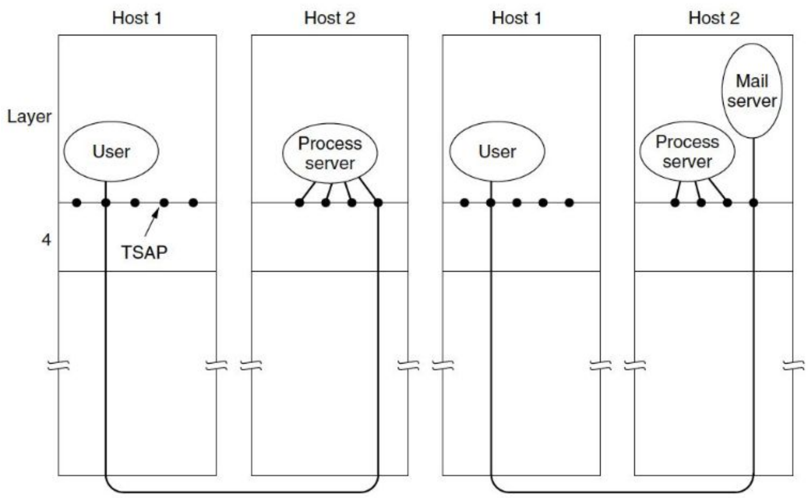
**4) Hoe werkt een leaky bucket en waarvoor staan de R en B?**
A leaky bucket is a way to limit how much data is in transit at any given time. Imagine a bucket with a hole in the bottom and capacity B. No matter the rate at which the water enter the bucket, the outflow is always at a constant rate R.When the bucket is full to capacity B, any additional water entering it spills over the sides and is lost. (Tanenbaum p.407-408)

**5) Teken voor een link aan 1Gbps hoe 1GB aan data verstuurd wordt voor:**
**a) B = 500MB, R=250Mbps**
Rechthoekige puls van 0 tot 4 seconden met hoogte R.
je moet hier ook rekening houden met B: eerst full speed tot bucket gevuld is en dan pas constante rate van R

**b)B = 0, R = 500Mbps**
Rechthoekige puls van 0 tot 2 seconden met hoogte R.

**6) Waarom gebruik je TSAP's?**
Transport Service Access Points (TSAPs) define an end-point for transport layer traffic, example ports. IP addresses (= example of NSAP) are not enough because multiple processes running on the same host may concurrently exchange data
Some servers may only operate sporadically. We can conserve resources via the **Initial Connection Protocol (ICP)**, it goes as follows:

1. Process Server listens for connections on well known TSAP addresses.
2. When a connection is received, the server process is started and passed the TSAP add




**7) Wat zijn de voordelen en nadelen van expliciete en impliciete signalen voor congestion control?**

- Explicit:
  - +: signals before congestion occurs so prevention method
  - +: faster than implicit
  - no extra packages need to be sent, complexity minimized with help of IP layer, optimal because will result in lower levels of loss and delay if implemented correctly, better for wireless connections (klopt niet: choke packets zijn een tegenvoorbeeld)
  - -: requires compatibility of both sending and receiving host for it to work.

    - -: extra packages might be sent over the already nearly-congested network
- Implicit:
    - +: Backward compatible: fix that's instantly deployable without changing packet formats
    - -: hear about congestion late, the network actually gets driven in congestion and then recovers fast, not ideal for wireless connection where packet loss can also occur due to poor connection.

**8) Define the Tinygram problem. What technique did we study to avoid this problem?**

The tinygram problem: what is the worst-case overhead of sending one char at a time using TCP one way on a 100% reliable connection?
We must consider both the IPv4 header and the TCP header as well as the data itself: - Transmission = 20 + 20 + 1 (bytes), Acknowledgement = 20 + 20 and window = 20 + 20 ====> 120 bytes overhead!
Techniques to avoid this problem: delayed acknowledgements: until timeout, wait for the data to be transmitted. If data is transmitted before time-out, piggyback acknowledgement in data, if timeout occurs, send acknowledgement.

- **Nagle's algorithm**, 1984 **(RFC 896)** while there is a sent segment with no ACK, buffer output until we have a full segment, then send at once.

```
if there is new data to send
  if the window size >= MSS and available data is >= MSS
    send complete MSS segment now
  else
    if there is unconfirmed data still in the pipe
      enqueue data in the buffer until an acknowledge is received
    else
      send data immediately
    end if
  end if
end if
```

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This wouldn't work well for computer games, real time systems...
Interacts badly with delayed acknowledgements. With both algorithms enabled, if Nagle's algorithm is being used by the sending party, data will be queued by the sender until an ACK is received. If the sender does not send enough data to fill the maximum segment size (for example, if it performs two small writes followed by a blocking read) then the transfer will pause up to the ACK delay timeout.

Open boek

**1) Real time protocol: a) Vergelijk de sequencing nummers van RTP en TCP. Waarom is er een verschil?**

RTP: 16 bit sequence number (Tanenbaum p.549)

TCP: 32 bit sequence number (Tanenbaum p.557)

TCP needs a larger sequence number because every byte on a TCP connection has its own sequence number (so we need a lot). In RTP the sequence number is just a counter for each RTP packet to detect lost packets. (correct?)

**b) Is RTP meer geschikt voor het downloaden van een video file dan TCP? Waarom?**

If the video file is real-time (streaming), then RTP is more suitable because it handles lost packets better (TCP will ask for retransmission but those aren't relevant anymore by the time they arrive). If the video file isn't streamed, but rather a 'normal' file transfer, TCP is better because of the congestion control, flow control and reliability it provides.

**c) Welk probleem kom je tegen bij een netwerk met high-volume RTP traffic, dat je niet hebt bij TCP?**

RTP doesn't have congestion control, TCP does. This means that high-volume RTP traffic can easily congest the network.

**2) Vergelijk TCP flow control met go-back-N. Geef verschillen en gelijkenissen.**

TCP is similar to GBN because both protocols have a limit on the number of unACK'd packets that the sender can send into the network. However, TCP is different from GBN because GBN requires the re-transmission of every unACK'd packet when packets are lost, but TCP only retransmits the oldest unACK'd one.

**3) Geef de verschillende manieren waarmee TCP omgaat met verschillende netwerken (congestie, sliding window, etc.).**

**4) Leg uit hoe een ACK clock werkt**

An ACK clock is a handy tool TCP uses to smooth out traffic and avoid unnecessary queues at routers. Imagine a network with a sender, a receiver and a router. There's a fast link in between the sender and the router, but a slow link in between the router and the receiver. The sender sends a burst of small packets to the receiver. Initially the four packets travel over the link as quickly as they can be sent by the sender. At the router they are queued while being sent because it takes longer to send a packet over the slow link. The receiver acknowledges the packets when they arrive. The times for the acknowledgements reflect the times at which the packets arrived at the receiver after crossing the slow link. The preserve this timing when they travel over the network. From these ACK's the sender can thus derive the rate that packets can be sent over the slowest link in the path. This is known as an ACK clock. If the sender injects new packets into the network at this rate, they will be sent as fast as the slow link permits, but they will not queue up and congest any router along the path. (Tanenbaum p.573-574)

**5) Gegeven output na PING message (zie labs). Welk sliding window protocol zou het geschiktste zijn in deze situatie?/ Gegeven een satellietverbinding tussen een ferry en een land. PING = +-2600ms. Hoe goed zullen de 3 geziene sliding window protocols (stop&wait, go-back-n, selective-repeat) hierop werken?**

**6) Delayed acknowledgements:**

**a) Verklaar en leg uit waarom deze nuttig zijn bij een betrouwbare verbinding zoals bv. TCP.**

Delayed acknowledgment is a technique used by some implementations of the Transmission Control Protocol in an effort to improve network performance. In essence, several ACK responses may be combined together into a single response, reducing protocol overhead.

**b) Zou dit goed samenwerken met Nagle's algoritme? Verklaar.**

No, If Nagle's algorithm is being used by the sending party, data will be queued by the sender until an ACK is received. If the sender does not send enough data to fill the maximum segment size of the receiver, the receiver will not send an ack and then the transfer will pause up to the ACK delay timeout.

(For example: consider a situation where Bob is sending data to Carol. Bob's socket layer has less than a complete packet's worth of data remaining to send. Per Nagle's algorithm, it will not be sent until he receives an ACK for the data that has already been sent. At the same time, Carol's application layer will not send a response until it gets all of the data. If Carol is using delayed ACKs, her socket layer will not send an ACK until the timeout is reached.)

**7) Waarom werkt UDP goed met DNS?**

DNS messages are short and simple messages. There is no need for a long connection (hence no TCP). (Tanenbaum p.622)

**8) Retransmission:**

**a) Wat zijn de voordelen van het implementeren van retransmission in de Transport Layer?**

Transport layer retransmissions avoid having to reimplement retransmission functionality in applications. Many applications can use the same transport protocol implemented in a library or in the operating system, reducing bugs, implementation time, etc.

**b) Waarom kan retransmission ook geïmplementeerd worden in de Data Link & Application Layer.**
De data link en application layer kunnen beiden ook controles uitvoeren om te checken of de transmissie al dan niet correct (bv checksum) en indien nodig een retransmissie vragen.

**9) Describe each phase of the three way handshake. List the state of the SYN and ACK bits during each phase.**
Phase 1: Both X and Y start in a closed state. It leaves that state when it does a passive open (LISTEN) or an active open (CONNECT). Suppose X want to connect. X sends a SYN=1, ACK=0 message to Y.
Phase 2: Y acknowledges the connection request (or declines the connection attempt). Y sends a SYN=1, ACK=1 message to X.
Phase 3: X sends a final acknowledgement to Y. The connection is now established. X sends a SYN=0, ACK=1 message to Y.

**10) Hoe gaat TCP om met heterogeniteit van netwerken en dynamische veranderingen in het netwerk?**

**11) Bereken de throughput van stop&wait bij een satellietverbinding tussen een ferry en een land met PING = +-2600ms als de maximum packet size = 1700B.**
**ping is RTT, dus slechts 1 keer ping duration voor het verzenden en aankrijgen van de ack?**
Stuur 1 packet, wacht op ack: 2 x ping duratie = 5200 ms per packet
=> 1700B / 5200ms = 326.92 B / sec

**12) Is packet loss een goeie indicator van congestion op een 802.11.weetikveel netwerk?**
No. A network must have very small levels of packet loss for this kind of congestion control to work. 1% is a moderate loss rate and by the time the loss rate reaches 10% the connection has effectively stopped working. However, for wireless networks such as 802.11 LANs, frame loss rates of at least 10% are common. This means that congestion control schemes that use packet loss as a signal will unnecessarily throttle connections that run over wireless links to very low rates.
Recap: To function well, the only packet losses that the congestion control algorithm should observe are losses due to insufficient bandwidth, not losses due to transmission errors. (Tanenbaum p.539)

**13) List the variables stored by the sender and receiver in: (Specify appropriate types for each of these variables)**

**(i.) stop-and-wait**

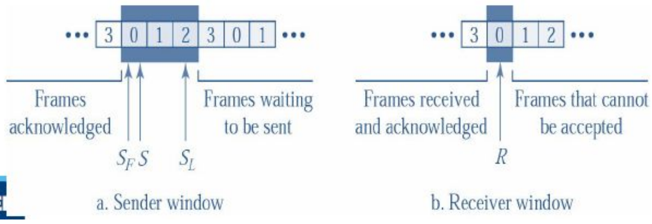
- Sender stores (**S**) with number of the last segment sent.
- Receiver stores (**R**) with number of next segment expected.

(slide 6.51)

**(ii.) go back n and**

## Go-Back-N: Variables

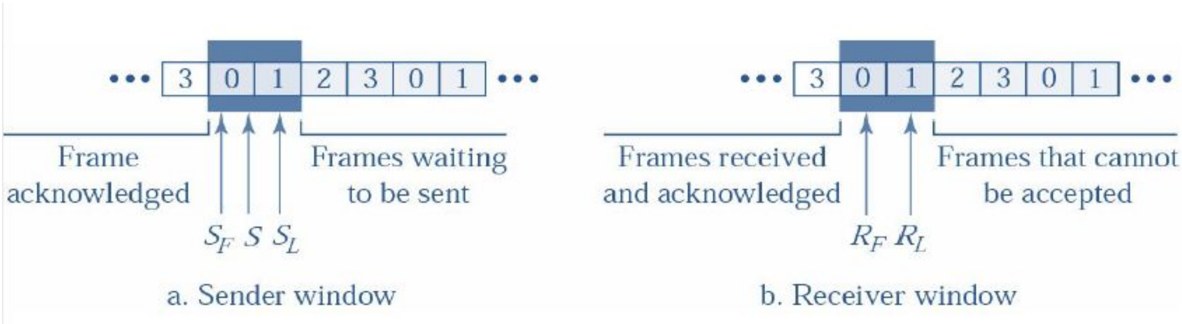
- S holds SEQ # of last segment sent.
- $S_F$ holds SEQ # of the first segment in window.
- $S_L$ holds SEQ # of the last segment in window.
- Receiver holds R: next expected SEQ #.




KU LEUVEN

(slide 6.61)

**(iii.) selective repeat.**




(slide 6.68)

Network Layer

# Gesloten boek

**1) Packet fragmentation:**

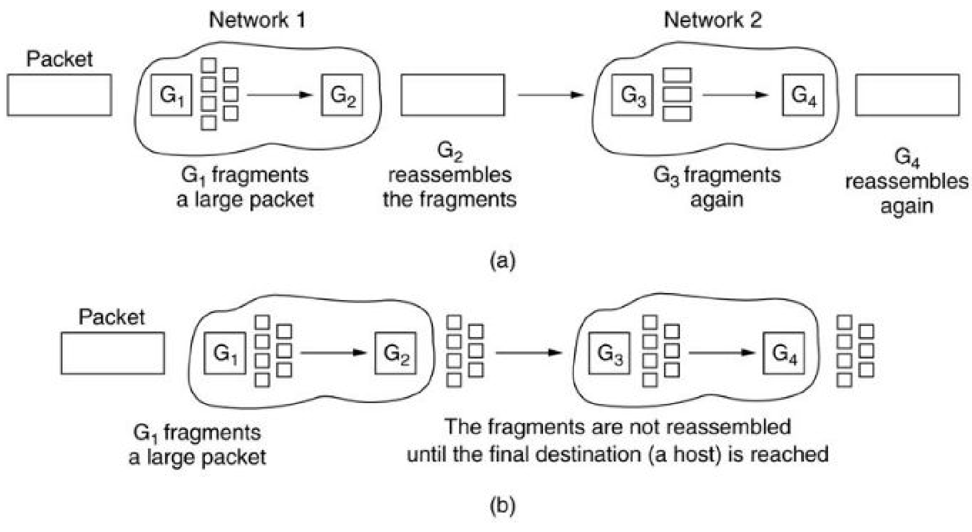
**a) Definieer fragmentation (network layer) en segmentation (transport layer)**

Fragmentation:

The process of breaking up packets into fragments and sending each fragment as a separate network layer packet. Fragmentation can be done before the packets are sent through the network layer (packet size is then determined by the ‘path maximum transmission unit’). Another option is letting routers break up packets into fragments.

There are two strategies for recombining the fragments back into the original packet, transparent or nontransparent.

a) Transparent:
   When packets are fragmented when being sent from router 1 to router 2 they are reassembled at router 2. Router 2 will then transmit the complete (reassembled) packet. Subsequent networks are not even aware that fragmentation had occurred.
b) Nontransparent:
   Refrain from recombining fragments at any intermediate router. Once a packet has been fragmented, each fragment is treated as though it were an original packet. Reassembly is performed only at the destination host.




Segmentation: This occurs during the original creation of the packets when a set of data doesn’t fit within the “Maximum Segment Size (MSS)”. The data is then chopped into multiple segments. MSS is determined during 3-way handshake.

**b) Waarom is er een verschil ("difference in approaches")?**

Segmentation is set up on beforehand (during the 3-way handshake), while fragmentation usually happens when dealing with networkheterogenity (the Maximum Transmission Unit is usually not known)

**c) Gaan packets door de network layer allemaal op dezelfde manier?**

No, this is the reason MTU (Maximum Transmission Unit) is unknown.

**d) Waarom kan je niet MTU en fragmentatie tegelijk gebruiken?**

Because packets using MTU have their header bits set to indicate that no fragmentation is allowed.

**e) Wat is de rol van ICMP bij MTU?**

MTU sends every packet with its header bits set to indicate that no fragmentation is allowed. If a router receives a packet that is too large, it generates an error packet (ICMP packet), returns it to the source and drops the packet. The sender can then reduce the packet size and try again.
(ICMP = Internet Control Message Protocol)

**2) Bespreek AODV in zo veel mogelijk detail. In welk geval zou AODV beter presteren dan OSPF?**

Ad-hoc On-Demand Vector routing is closely related to distance vector routing. Routes are only discovered when someone wants to send a datagram (i.e. reactive, not proactive). This is better for highly dynamic networks.

Route discovery:
Broadcasts a ROUTE_REQUEST message which contains a TTL and SEQ number. All neighbors (routers within range) decrement TTL with one and repeat broadcast. Initially ROUTE_REQUEST is sent with TTL = 1, the TTL is incrementally increased until a route is found.
Discovered node sends ROUTE_REPLY back along the path of the incoming ROUTE_REQUEST. A hop counter is incremented at each intermediate hop and intermediate nodes add the route to their tables.

Since NO_ROUTE exceptions have very long time-outs, a lot of traffic (inefficient) when node failure is common. If a node detects one of its neighbors is down, it removes every path using this node from its routing table and notifies neighbors using these routes to do the same.

**3) Leg Round Robin fair queuing uit. Hoe kan een host hier misbruik van maken?**

Way of packet scheduling, maintains a separate queue for each flow, router sends a packet from each queue in turn. Can be abused by sending very large packets.

**4) Geef 2 vd 3 mogelijkheden voor congestion notification + voor- en nadelen.**
Choke packets are explicit notifications provided by congested routers.

- A datagram is sent to the sender.
- The packet is marked so that no further choke packets will be generated later on => causes sending entity to throttle send rate.

Disadvantage: extra load on an already (near) congested network.

ECN (Explicit Congestion Notification): used to tag packets as they travel through routers.
ECN-tagged packets reach destination host, which messages the sender to throttle send rate.
ECN has lower overhead than choke packets, but takes longer to enact change.

RED (Random Early Detection): packet loss is a slow indicator. RED will drop random packets as congestion approaches. This idea is counter-intuitive, but has advantages:

- makes packet loss a more reliable congestion signal
- reduces the delay between congestion starting and transport-layer throttling

It's a good way to signal fast senders because fast senders tend to have many more packets in the queue so randomly dropping packets tends to affect them more.
This also reduces book-keeping overhead.

**5) Geef een volledige definitie van unicast, anycast en broadcast.**
Unicast: one-to-one
Anycast: one to any of many
Broadcast: one-to-all

| TYPE | ASSOCIATIONS | SCOPE | EXAMPLE |
|---|---|---|---|
| Unicast | 1 to 1 | Whole network | HTTP |
| Broadcast | 1 to Many | Subnet | ARP |
| Multicast | One/Many to Many | Defined horizon | SLP |
| Anycast | Many to Few | Whole network | 6to4 |

**6) IPv6:**
**a) Leg het autoconfiguration-proces uit.**

Every interface that is Ipv6 enabled has a link local address that uses the MAC address of the interface as a basis for the construction. With IPv6 we don't need a DHCP, it gets an IP address via SLAAC (Stateless Address Autoconfig) this uses the Neighbor Discovery Protocol to find what subnet it belongs to as it talks to the local router with a Router Solicitation and receives RA (the "local link unicast"). Now the device knows the subnet, it creates a host address associated with the interface's MAC address. It will send out a Duplicate Address Detection to make sure nobody on the network is already using this address.
Autoconfiguration on the basis of the network card MAC address is therefore a particular privacy concern for mobile devices, such as laptops, because when they access the Internet from different LAN, their MAC based interface identifier would always stay the same. Thus the MAC address based interface identifier can be used to track the movement and usage of a particular mobile device

**b) Waarom kan autoconfiguration niet op IPv4 toegepast worden?**

**7) Beschrijf de relatie tussen Link State Routing, BGP en OSPF. Hoe interageren deze protocollen?**
link state routing
process:

1. Discover its neighbors network addresses
   On boot, each router sends a HELLO packet to all neighbors, all routers who receive will respond witch datagram containing it's address.
2. Set the link cost to all neighbors
   Link costs can be distance, signal strength... Most common: measure inversely proportional to bandwidth or delay.
3. Construct a datagram with all learned
   A packet is built containing: (a) address of sender, (b) sequence number, (c) age and (d) a vector of all costs to each neighbor.
   Each router must make decisions based upon local knowledge.
4. Distributing datagrams (Send packet to all other router)
   Datagrams are distributed using flooding: forward all incoming packets on all links except the one it arrived on. 32bit Seq. numbers are used by routers to discard packets already seen.
   To recover from crashes, age of packets is also included. Packets older than some threshold are discarded.
5. Compute shortest path to all other routers using Dijkstra's Algorithm
   Once all link state packets have been received the router can reconstruct the whole graph using Dijkstra's algorithm on the graph to find shortest path to each remote router. Note: every link is represented twice, once for each direction as costs may be asymmetric.

The major difference between OSPF and BGP is that the OSPF is an intradomain routing protocol while BGP is the interdomain routing protocol. The OSPF protocol uses link state routing. On the other hand, BGP protocol uses path vector routing.
OSPF: Easy to implement, supports load balancing and hierarchy, fastest route(s) preferred,  UDP
BGP: complex to implement, slow convergence, for large scale networks, connect the AS's (the internet), best path algorithm,  TCP, internal details of AS are hidden

**8) Bespreek en geef verbanden: Dijkstra, Bellman Ford, Link state routing en BGP.**
Link State en bgp => zie vorige
Dijkstra: algorithm to find shortest path between nodes in a network, maintain a table and set distance to all nodes infinity, examine neighbors and re-label distance, next node to examine is the node with the shortest total distance.
Used in distance-vector routing protocols, eg RIP (Routing Information Protocol).
Process:
1) Each router measures distance (e.g. delay) to each neighbor which is measured using special ECHO datagrams.
2) Each router periodically exchanges its routing tables with all neighbors.
3) When a node receives distance tables from its neighbors, it calculates the shortest routes to all other nodes and updates its own table to reflect any changes.
Disadvantages:

- It does not scale well.,
- Introduces count to infinity problem: path length can only increase by 1 each round and no router is aware if it itself is included in a path.
- Good news travels fast, but bad news travels slowly.

=> vector routing gets replaced by link state routing

**9) Hoeveel IP-adressen moeten een router en een switch hebben?**
Router: 1 IP-address per interface: Creates a separate LAN per interface "port". Devices connected to different ports can communicate through the routers. Each interface has its own ip address and network ID of course.
Switch: none, can have an IP-address for remote configuration purposes.

## Open boek

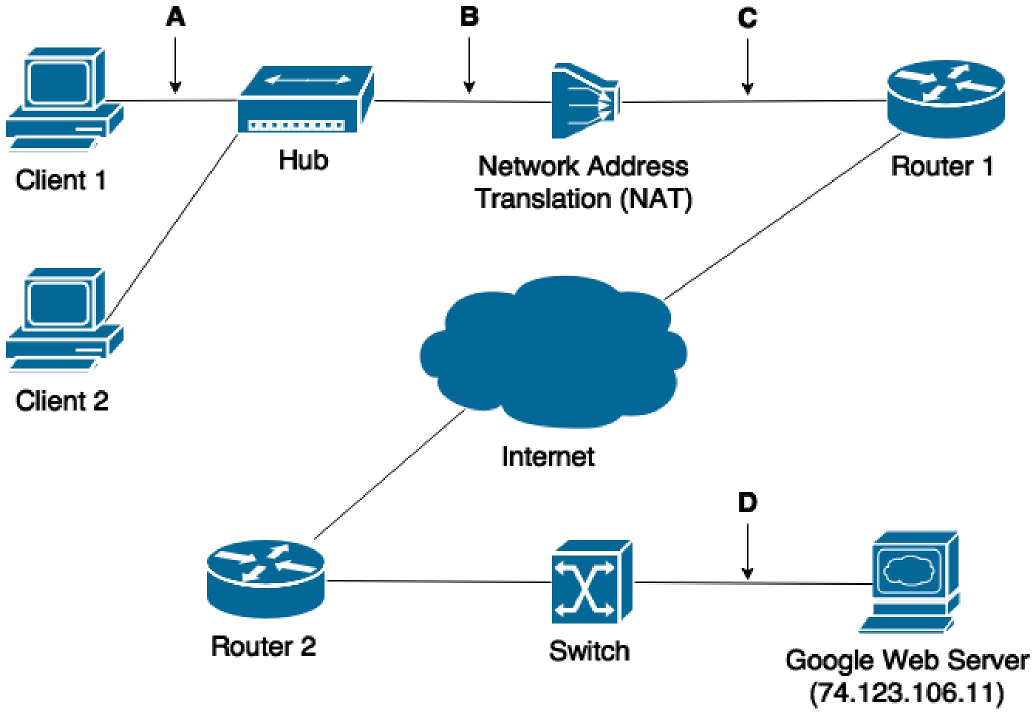
**1) Zeg wat fragmentation is en leg het belang van ICMP packets uit om dit op te lossen.**

**2) a) Bespreek waarom IPv6 met automatische netwerkallocatie onveilig is.**
Dit gebeurt adhv het MAC adress vh apparaat. Een mobiel apparaat zal dus op verschillende plekken hetzelfde IPv6 address hebben, op deze manier kan men zien waar dit apparaat overal komt.

**b) Waarom is er geen checksum in IPv6?**

**c) Gegeven een IPv6 address. Geef de functies van de verschillende delen en kort het adres af indien mogelijk.**

**3) The following network has 100 Mbps Ethernet links and follows the Tanenbaum stack:**

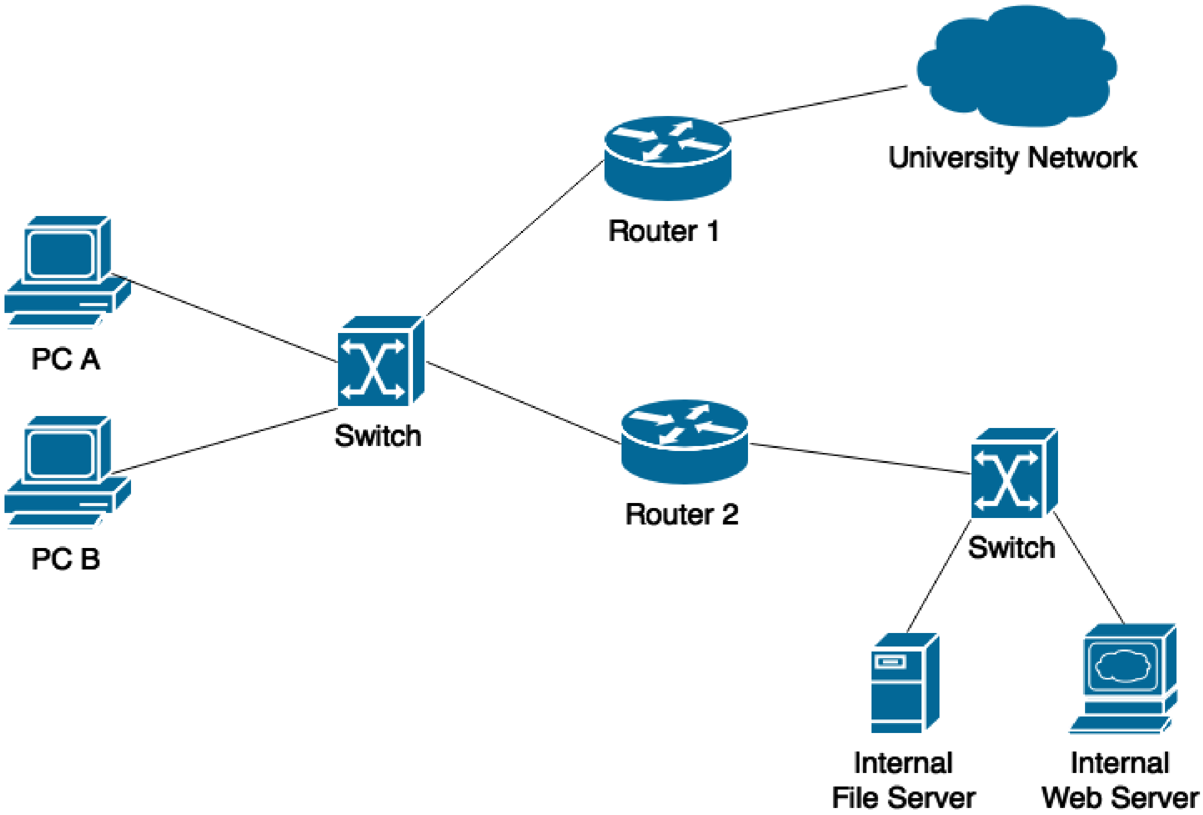



**a) Assign meaningful addresses to all network structures. Which are private addresses and which are public?**

**b) Client 1 accesses Google. Give the Protocol Data Units on the four points A, B, C and D in the network (only towards Google, not in the opposite direction). Include the TSAP, NSAP and data link address and any header content you think is important.**

**c) Your ISP only assigns one public IPv4 address to your home. Would it be possible to replace the NAT box in your home with a router?**
No because then every device in your home would need a public IPv4 address.

4) The block 192.168.22.0/24 is assigned to this network:




a) Assign appropriate addresses to all network structures.

b) Give the routing tables entries of PC A, router 1 and router 2 so they can access the university network, the internal file server and the internal web server.

c) Give the ARP cache of PC A, router 2 and the internal web server when PC A accesses the internal web server.

5) The IP header includes a TTL field: a)Why can't you have a TTL field in Ethernet?

b) Can you replace the TTL field in the IP header with a similar field in the TCP/UDP header?

6) Distance Vector Routing: a) Leg kort de werking uit.

b) Welke protocollen maken hier gebruik van en hoe gaan ze met het count-to-infinity probleem om?
BGP
AODV

**7) Congestion Notifications: Vergelijk choke packets, ECN en RED op vlak van:**
**a) efficiëntie**

**b) snelheid van het effect**
Choke packets are faster than ECN and ECN is faster than RED.

**c) invloed op het netwerk/effectiviteit**

**8) Adressering: Vergelijk adressering in Network Layer met die in Data Link en Application Layer. Geef aan waarom dit nodig is. Bespreek uitvoerig.**

**9) Applications of the Internet Protocol: In your opinion, should the Internet Protocol be used on low power Internet of Things networks such as IEEE 802.15.4 (i.e. the physical layer of TSMP and BMAC)?**
- **List key advantages and disadvantages of using IP on IEEE 802.15.4.**
- **Is IPv6 better suited to IoT devices than IPv4? Justify your answer.**

## Data Link Layer

## Gesloten boek

**1) Hidden/exposed terminal problem:**
**a) Wat is het hidden terminal problem?**
The hidden terminal problem is due to the fact that a node (say A) transmitting to another node (say B) cannot hear transmissions from another node C, which might also be transmitting to B, and might interfere with the A-to-B transmissions.

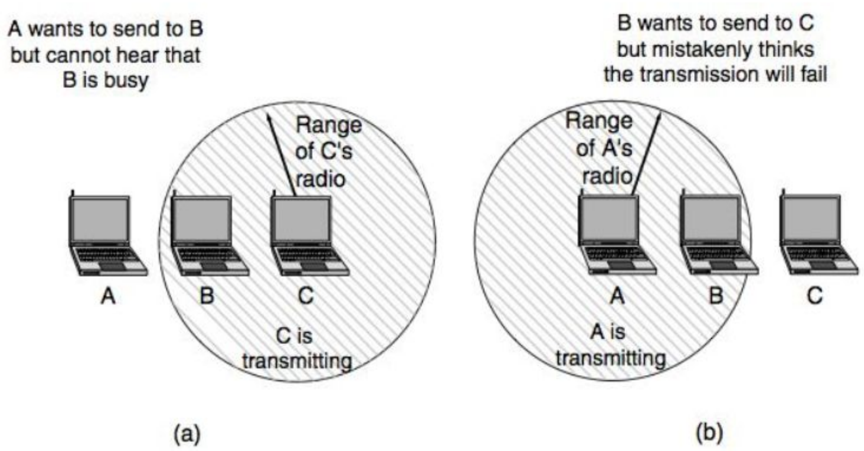



**Figure 4-26.** (a) The hidden terminal problem. (b) The exposed terminal problem.

**b) Hoe wordt dat opgelost door MAC protocols (geef 2 voorbeelden)?**
MACA: Multiple Access with Collision Avoidance: if a node wants to send, it sends a Request To Send (RTS) with the length of the frame to send, then B replies with a Clear To Send (CTS) also containing the length, in doing so all nearby stations can also detect this transmission and know the channel is busy for the time it takes to transmit the frame. (Tanenbaum p.279)
CSMA/CA (Carrier Sense Medium Access with Collision Avoidance): Station always starts with a random back-off, if the channel is idle, it sends and waits for an ACK from the receiver, advantages: back-off helps avoid collisions and the ACK's are used to infer collisions as they cannot be detected. Uses virtual channel sensing based on NAV: each frame carries NAV field that says how long the sequence will take to complete (Tanenbaum p.303-305)

**c) Wat is het exposed terminal problem?**
C doesn't send because it senses the interference from B. Causes unnecessary delay.

**d) Kan CSMA hierbij helpen?**
Nope. What matters for reception is interference at the receiver, not at the sender. CSMA merely tells the us whether there is activity near the transmitter by sensing the carrier.(Tanenbaum p.278)

**e) Wat zijn de voordelen van channel hopping?**
Interference tends to cluster around a subset of channels, so if you pick a single channel with an interference source, transmission rate will be really bad. Each available frequency band is divided into sub-frequencies. Signals rapidly change ("hop") among these in a predetermined order known to both transmitter and receiver. Interference at a specific frequency will only affect the signal during that short interval.

**2) Beschrijf pure ALOHA in zo veel mogelijk detail. Wat is de verbetering van slotted ALOHA.**
Pure ALOHA: user can transmit upstream frames whenever they like, if collisions occur resulting in garbage frames, this is detected by the user via downstream retransmission of all received packets. If the frame was lost or collision occurred, wait random interval (synchronous would keep colliding) and retransmit.
Slotted: requires synchronisation, time is divided into slots and nodes can only send at beginning of a time-slot => improves performance.

**4) Leg uit hoe B-MAC werkt en waarom het over TSMP zou worden gebruikt.**
B-MAC: based upon a CSMA scheme, no time-synchronisation so joining network is quick and low-cost, uses Low Power Listening (LPL): nodes only listen for short

sampling periods, if traffic is detected the radio remains active and listens to receive packet, senders use preamble (with) to make sure the radio is active when the data arrives.
Offers link-layer ACK's (optional), offers clear channel assessment: allows for low power testing of whether a channel is active, highly configurable, decentralized (good for scaling), well suited for unpredictable traffic (unlike TSMP).
Disadvantage: loss energy for unnecessary listening and radio use (preamble).

**5) Leg uit hoe TSMP werkt en waarom het over B-MAC zou worden gebruikt.**
TSMP = Time Synchronised Mesh Protocol
- For many-to-one data gathering
- uses centralized manager (master node) for synchronisation: upon joining the network, node listens for transmission and uses this timing to set own clock, listens through complete time-cycle to discover when to transmit and listen => radio off for the rest of the time => high cost in (re)joining network.
- Uses Time Division Medium Access (TDMA) => min. Collisions due to simultaneous transmission => less retransmissions => less power consumption.
- Precise QoS: time-slots can be precisely allocated, higher demand nodes can get multiple time slots.
- Avoids interference with channel hopping (zie vorige vraag).
- Reliability ensured via link-layer ACK's at each hop.
- Each node has multiple parents => spatial redundancy (try different node) and temporal redundancy (try different time).
- Packet is 128 bytes, with 80 payload.

Disadvantage: less robust due to and high load on manager, not scalable: TSMP packet only allows 1 byte for ID's, high latency for sporadic traffic.

**6) Compare Slotted ALOHA and TSMP on time synchronization. Give similarities and differences, and advantages and disadvantages.**
Similarities:
- Stations have to be synchronised.
- They try to minimize collision.
- One central station to send data to.

**7) Give a scenario where BMAC outperforms TSMP, and one where TSMP outperforms BMAC. Consider traffic and environmental characteristics.**

**8) Imagine a 'Null MAC' protocol, where the node listens 100% of the time, sends immediately whenever a message is ready and does no retransmissions: a) What problems do you think would occur in a network with high traffic?**

Lots of collisions -> Messages don't arrive because there are no retransmissions.
**b) What problems do you think would occur in a crowded place such as a city center?**

**9) TSMP: Leg gedetailleerd uit.**

**10) Switches - Hubs: Leg uit wat het is. Wat zijn de voordelen van een switch t.o.v. een hub?**
Hub: device where all the ethernet lines are joined together, addresses the "christmas light" problem: makes it easier to debug networks, if a link is broken it only affects one host. Doesn't increase capacity.
Switch: high-speed backplane that connects hosts on a single line, only outputs frames to destination ports (=> improves capacity), usually full duplex => no more CSMA/CD unless it has multiple hosts connected via a hub
=> improves performance since send and receive can occur sim.
=> impact promiscuous mode is limited
More expensive because it needs buffers if 2 input ports try to send frames to same output port.

**11) Ethernet: a) Leg uit waarom er een beperking is bij klassieke Ethernet in kabellengte.**
Zie tekening boek
The signal used to send information in classic Ethernet is Manchester encoding. This signal is susceptible to noise when sent over a wire. After it has travelled a certain distance, the signal is not recognisable anymore, hence a maximum cable length.

**b) Is deze hetzelfde voor Fast Ethernet met hub?**
No, reduction of bit transmit time leads to cut of cable length by factor of 10 (2.5km to 250m).

**c) Is deze hetzelfde voor Fast Ethernet met switch?**
No since on a switch (with duplex cables) CSMA/CD is not used => cable length only limited by signal strength.

**d) Waarom moet je padden, geef twee gevallen en bespreek.**
Padding is extra bytes that are added at critical places within a protocol so that fields are aligned conveniently. UDP checksum, the 4 unused bits following TCP header length.

**12) Kan het Pure ALOHA algoritme ook werken op 802.11? Verklaar.**

**13) Neem aan dat ALOHA werkt op een radio-medium met n frequency channels, beschrijf hoe frequency hopping aan de protocol toegevoegd kan worden, haal inspiratie uit de frequency hopping benadering van TSMP.**

## Open boek

**1) Er worden 2 situaties van low power listening gegeven, geef voor beide een protocol dat er het best bij past: a) een systeem waarbij sensoren vrije plaatsen in een parking meten**

**b) sensoren aan vaten met chemische stoffen, die niet bij elkaar in de buurt mogen komen, anders starten ze een alarm.**

**2) Zou packet loss goed zijn om congestion aan te duiden in een 802.11 netwerk?**

**3) Leg uit waarom byte stuffing werkt. Geef een numeriek voorbeeld waar bit stuffing efficiënter is dan byte stuffing.**

**4) Stel dat hubs en switches evenveel kosten. Geef een situatie waarbij een hub beter zou zijn dan een switch.**

**5) Combining BMAC and TSMP: Propose a scheme to usefully combine the Low Power Listening (LPL) approach of the Berkeley Medium Access Control (BMAC) protocol and Time Synchronised Mesh Protocol (TSMP). Your proposed scheme should allow for the seamless coexistence of the both protocols.**

**6) Long Range Low Power Networks: LoRa is a new network technology that is gaining popularity in the Internet of Things due to its support for long range and low power communication. We did not study the LoRa protocol in class, however you should no be able to use your knowledge of data link protocol design to speculate about its performance. Here are the important protocol details: ● LoRa radios have a range of approximately 20 KM.**
**● In the worst case, a LoRa packet takes over 1 second to send.**
**● LoRa has no notion of network time, nor time synchronisation.**
**● Any user may deploy a LoRa network gateway without a license or other usage restrictions.**
**What problems would you expect the LoRa network to experience in real-world deployments? Justify your answer.**